Question 3

_After digitization process a digital image with M rows and N columns and L, intensity levels. the number of bits required to store this image is

a. M*N*log2(L)

b. M*M*L

c. M*N*\

Question 4

_An image is defined as a function of f(x,y), where x,y represents .....and f represent.....

a. intensity , resolution

b. spatial coordinates , amplitude

c. brightness , position

Question 5

_Bit plane slicing is a method of representing an image with different planes which plane represent the most significant data. Show its content for pixel (3,7) in the following image matrix.

```
[255, 210, 202, 205, 204, 255, 255, 255, 253, 255,
 253,  42,   2,   1,   0, 252, 254, 253, 255,  96,
 254,  49,   1,   0,   3, 255, 254, 253,  83,   0,
 255,  51,   0,   0,   0, 255, 253,  76,   0,   1,
 254,  47,   0,   6,   0, 255,  45,   0,   0,   0,
 255,  54,   0,   0,   2, 249, 143, 155, 155, 155,
 255, 255, 255, 255, 255, 255, 143, 155, 155, 155]
```

a. Least significant, 0

b. non of the mentioned

c. Most significant, 1

Question 6

_Digital image processing:

a. It describes the ratio between the maximum and minimum intensities in the Image

b. It is Processing of image data for storage, transmission and representation for autonomous machine perception

c. It is a graph, which gives you an overall idea about the intensity distribution of an image

Question 7

_For a continuous image f(x, y), Quantization is defined as

a. None of the mentioned.

b. Digitizing the coordinate values

c. Digitizing the amplitude values

Question 8

_For a continuous image f(x, y), sampling is defined as

a. None of the mentioned.
b. Digitizing the amplitude values
c. Digitizing the coordinate values

Question 11

_the first and foremost step in Digital Image Processing is

a. Image acquisition
b. Image rotation
c. Affine transformation

Question 14

_The method which increases the low dynamic range of the gray levels in a low contrast image to cover the full range of gray levels in order to improve image contrast is.

a. Contrast stretching
b. Histogram equalization
c. Image smoothing

Question 15

Which of the following represent transformation function of histogram equalization

a. S=(L-1)*CDF(r)
b. S = c * log (1 + r)
c. S=L-r

Question 16

_Which of the following used to enhance the problem of shifting an object's location?

a. Image smoothing
b. Image translation
c. Image skewing

_The following image matrix represents the pixel values of an 8-bit image. Which of the following represents the D_m distance between pixels p (1,2) and q (4,6), given the connectivity set ξ={0,1,2,3,6}

```
[255, 210, 202, 205, 204, 255, 255, 255, 253, 255,
 253,  42,   2,   1,   0, 252, 254, 253, 255,  96,
 254,  49,   1,   0,   3, 255, 254, 253,  83,   0,
 255,  51,   0,   0,   0, 255, 253,  76,   0,   1,
 254,  47,   0,   6,   0, 255,  45,   0,   0,   0,
 255,  54,   0,   0,   2, 249, 143, 155, 155, 155,
 255, 255, 255, 255, 255, 255, 143, 155, 155, 155]
```

a. 4
b. 2.5
c. 3
d. non of the mentioned

_The following image matrix represents the pixel values of an 8-bit image. Which of the following represent N4 and ND neighbors for pixel (3,4), respectively?

```
[255, 210, 202, 205, 204, 255, 255, 255, 253, 255,
 253,  42,   2,   1,   0, 252, 254, 253, 255,  96,
 254,  49,   1,   0,   3, 255, 254, 253,  83,   0,
 255,  51,   0,   0,   0, 255, 253,  76,   0,   1,
 254,  47,   0,   6,   0, 255,  45,   0,   0,   0,
 255,  54,   0,   0,   2, 249, 143, 155, 155, 155,
 255, 255, 255, 255, 255, 255, 143, 155, 155, 155]
```

a. {0,2,6,255}-{0,255,0,249}

b. {3,0,0,255}-{0,255,6,255}

c. {255,249,0,45}-{0,253,3,143}

_For the following image matrix, which of the following represent the enhancement effect at pixel (5,1) which used to enhance white or grey detail embedded in dark regions of an image.

```
[255, 210, 202, 205, 204, 255, 255, 255, 253, 255,
 253,  42,   2,   1,   0, 252, 254, 253, 255,  96,
 254,  49,   1,   0,   3, 255, 254, 253,  83,   0,
 255,  51,   0,   0,   0, 255, 253,  76,   0,   1,
 254,  47,   0,   6,   0, 255,  45,   0,   0,   0,
 255,  54,   0,   0,   2, 249, 143, 155, 155, 155,
 255, 255, 255, 255, 255, 255, 143, 155, 155, 155]
```

a. non of the mentioned

b. 51

c. 3

_For the following image matrix, which of the following represents the enhancement effect at pixel (2,3) used to enhance an image with low brightness?

```
[255, 210, 202, 205, 204, 255, 255, 255, 253, 255,
 253,  42,   2,   1,   0, 252, 254, 253, 255,  96,
 254,  49,   1,   0,   3, 255, 254, 253,  83,   0,
 255,  51,   0,   0,   0, 255, 253,  76,   0,   1,
 254,  47,   0,   6,   0, 255,  45,   0,   0,   0,
 255,  54,   0,   0,   2, 249, 143, 155, 155, 155,
 255, 255, 255, 255, 255, 255, 143, 155, 155, 155]
```

a. 51

b. 32

c. 0